ECE 501 Project Week 7 Report

Project Title: Vehicle Orientation Estimation Using Conventional Image Processing

Group Number: 14

Project number: 8

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Github Repository Link: https://github.com/hetpatel5107/DIP_2026_Project8_Group14

- **Introduction**

This week we explored a new method, Principal Component Analysis(PCA) as another approach to estimating vehicle orientation from aerial images. This method works by analysing the distribution of edge pixels and finding the direction in which they are spread the most. We first tested PCA on a single vehicle crop to understand how it estimates the vehicle's main axis. After that, we applied the same method to 240 vehicle crops from five aerial images. We then calculated the errors overall and across different vehicle classes to understand how well PCA performs.

- **PCA Method**

Theory: PCA (Principal Component Analysis) is a mathematical algorithm used to find the primary direction in which a set of data is spread out the most. The first step in estimating the orientation of a vehicle is to use the Canny edge detection method to determine all of the pixels representing the outline of the car. Now we represent each individual edge pixel as (x,y) the data point on a huge 2-D scatter plot.

This is done by PCA by computing the covariance matrix of these coordinates to calculate the covariance between x and y:

$$\Sigma = \frac{1}{n-1}\sum_{i=1}^{n}(P_i - \mu)(P_i - \mu)^T$$

(Where P_i represents the coordinates of a specific edge pixel, and μ *is the mean center point of all the pixels).*

PCA calculates “eigenvectors” (directions) and “eigenvalues” (how much the spread has in each direction). The most spread out direction of the pixels is the direction of the eigenvector corresponding to the largest eigenvalue. Since a vehicle is longer than it is wide, this is by nature the direction of maximum variance for a vehicle's front-to-back orientation axis.

- **PCA Implementation**

Code:

```
single_axis_angle, single_center, single_vector =
estimate_pca_axis(single_edges)

single_estimate = convert_to_0_90(single_axis_angle)

height, width = single_gray.shape
line_length = max(height, width)

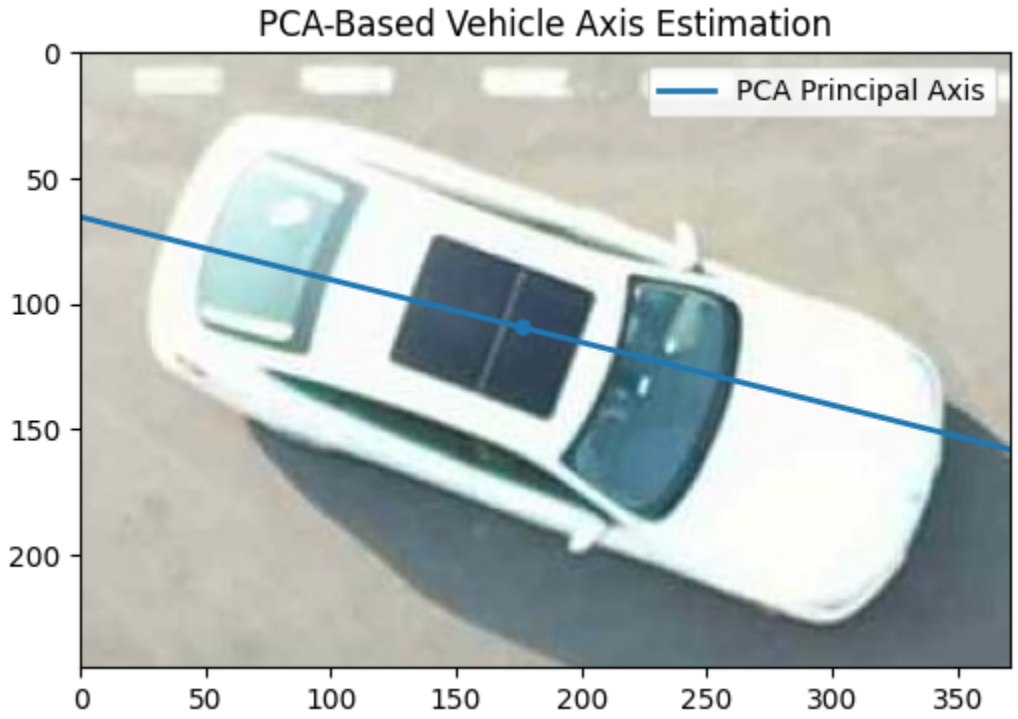
x1 = single_center[0] - line_length * single_vector[0]
y1 = single_center[1] - line_length * single_vector[1]

x2 = single_center[0] + line_length * single_vector[0]
y2 = single_center[1] + line_length * single_vector[1]

plt.figure(figsize=(6, 6))
plt.imshow(single_image[:, :, ::-1])
plt.plot([x1, x2], [y1, y2], linewidth=2, label="PCA Principal Axis")
plt.scatter(single_center[0], single_center[1], s=25)
plt.xlim(0, width)
plt.ylim(height, 0)
plt.title("PCA-Based Vehicle Axis Estimation")
plt.legend()
plt.show()

print("PCA Axis Angle:", round(single_axis_angle, 2), "degrees")
print("PCA Estimated Axis (0-90):", round(single_estimate, 2), "degrees")
```

Output:




```
PCA Axis Angle: 14.03 degrees
PCA Estimated Axis (0-90): 14.03 degrees
```

Interpretation:

The output image shows one vehicle with a line drawn on it showing the estimated principal axis. The direction of maximum variance has been identified well by the PCA algorithm, which treated the Canny edge-pixels as a 2D scatter plot. The estimated axis being physically longer than the vehicle itself, aligns well with the vehicle's true front to back orientation, which shows that PCA can be used to achieve a good estimate of a vehicle's heading direction solely based on its spatial edge geometry.

- **Overall Error Distribution**

Code:

```
print("\nResults for every vehicle crop:")

print(results_df.to_string(index=False))



mae = results_df["Error (degrees)"].mean()

minimum_error = results_df["Error (degrees)"].min()

maximum_error = results_df["Error (degrees)"].max()



print("\nOverall PCA Statistics")

print("Number of crops:", len(results_df))

print("Mean Absolute Error:", round(mae, 2), "degrees")

print("Minimum Error:", round(minimum_error, 2), "degrees")

print("Maximum Error:", round(maximum_error, 2), "degrees")



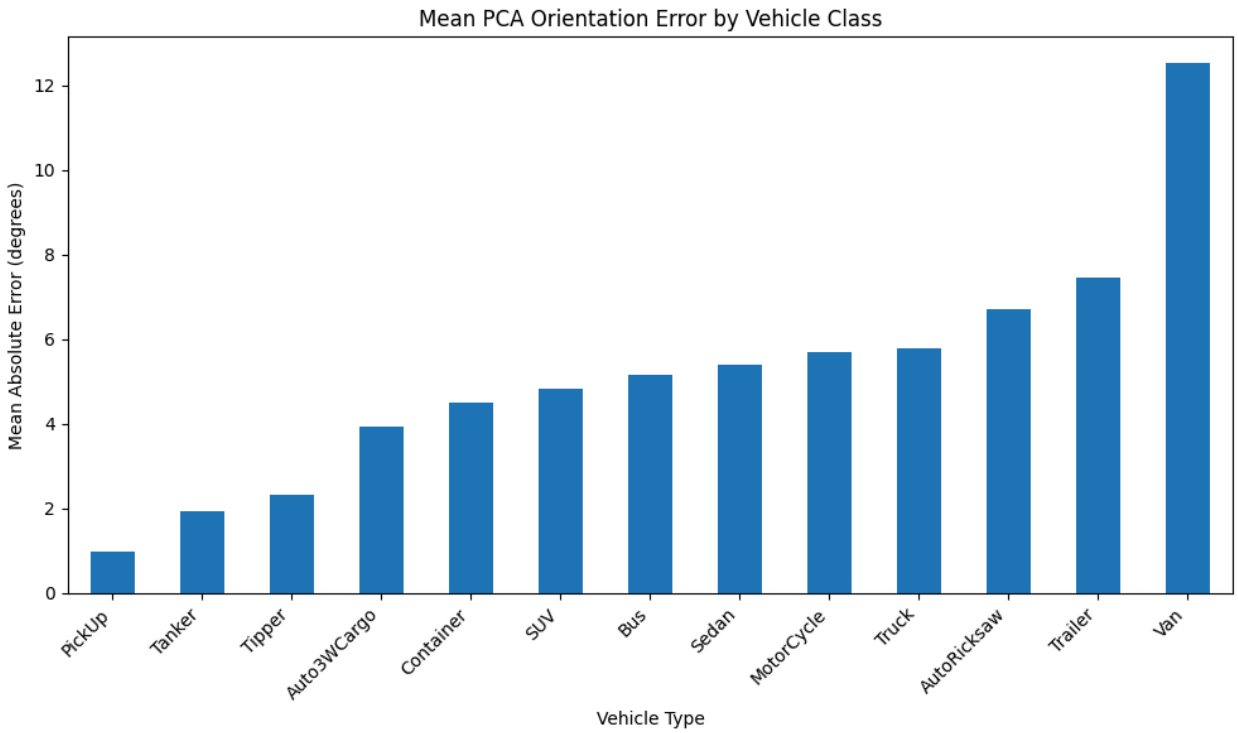
results_csv = os.path.join(output_dir, "Week7_PCA_Results.csv")

results_df.to_csv(results_csv, index=False)



print("Saved results table:", results_csv)
```

Output:



Interpretation:

In this graph, the mean absolute orientation errors of PCA for various vehicle classes are compared with each other. Pickup trucks exhibit the lowest mean value of approximately 1 degree while tankers and tippers have a mean error of about 1 degree as well, which indicates better orientation estimation of these vehicle classes. However, vans exhibit the highest mean value of about 12.5 degrees and thus have low accuracy. In addition, trailers and autorickshaws also exhibit relatively higher mean values as compared to most other classes. Other vehicle classes such as SUVs, buses, sedans, motorcycles, and trucks have mean values of around 4 to 6 degrees. It can be seen that there is variability in the accuracy of PCA with respect to vehicle classes.

- **Mean Error by Vehicle Class**

Code:

```
class_stats = (
    results_df.groupby("Type")["Error (degrees)"]
    .agg(["count", "mean", "median", "min", "max"])
    .sort_values("mean")
)

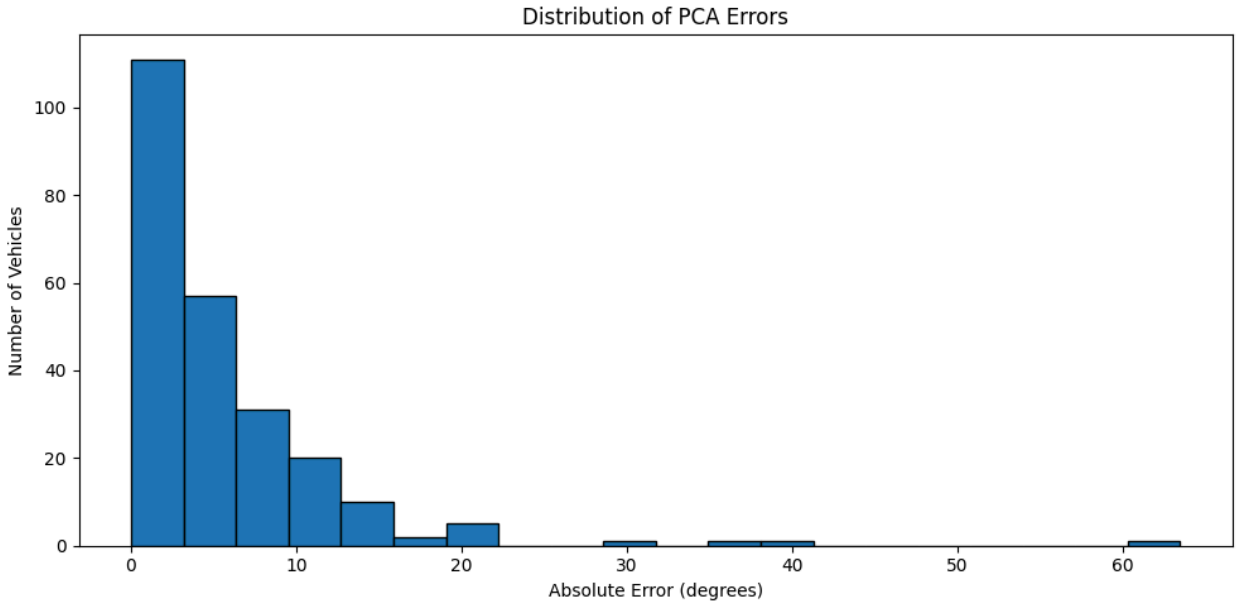
print("\nPCA Error Statistics by Vehicle Class:")
print(class_stats.round(2).to_string())

class_stats_csv = os.path.join(output_dir, "Week7_PCA_Error_By_Type.csv")
class_stats.round(2).to_csv(class_stats_csv)

print("Saved vehicle-type summary:", class_stats_csv)

plt.figure(figsize=(10, 6))
class_stats["mean"].plot(kind="bar")
plt.xlabel("Vehicle Type")
plt.ylabel("Mean Absolute Error (degrees)")
plt.title("Mean PCA Orientation Error by Vehicle Class")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.savefig(os.path.join(output_dir,
"PCA_Mean_Error_By_Vehicle_Class.png"), dpi=200)
plt.show()
```

Output:



Interpretation:

The histogram represents the absolute errors in the orientation estimates obtained by PCA relative to all the sampled vehicles. It can be observed that the majority of the vehicles have an error less than 10 degrees, which means that PCA performs quite good for the orientation estimation of the majority of the vehicles. With the increase in the absolute value of the error, the number of vehicles also declines and leads to the formation of a right-skewed distribution. It is observed that only a few vehicles have an error above 15 degrees, while a few of them exhibit a much higher value of the error, around 63 degrees.

- **Per-Class Error Analysis**

Code:

```
results_df.round(2).to_csv(
    os.path.join(output_dir, "Week7_PCA_Results.csv"),
    index=False
)

class_stats.round(2).to_csv(
    os.path.join(output_dir, "Week7_PCA_Error_By_Type.csv")
)
```

Output:

Type	count	mean	median	min	max
PickUp	5	0.97	0.55	0.03	2.76
Tanker	2	1.94	1.94	0.71	3.17
Tipper	5	2.31	2	0.97	4.76
Auto3WCargo	1	3.94	3.94	3.94	3.94
Container	2	4.52	4.52	0.88	8.16
SUV	55	4.82	3.11	0.02	20.08
Bus	1	5.16	5.16	5.16	5.16
Sedan	16	5.4	4.5	0.61	13.65
MotorCycle	114	5.69	3.7	0.07	38.84
Truck	5	5.8	4.38	0.91	12.95
AutoRicksaw	25	6.73	4.84	0.03	30.16
Trailer	1	7.47	7.47	7.47	7.47

Van	8	12.52	3.61	0.78	63.46

Interpretation:

1. The dataset consists of 240 samples of cars belonging to 13 classes.
2. There are most samples for the class of motorcycles – 114, and for the class of SUV – 55.
3. Pick-up truck is the class that possesses the smallest value of the mean error – 0.97°.
4. The maximum mean error is 12.52° and belongs to the class of vans.
5. The mean errors of motorcycles and SUV are 5.69° and 4.82° correspondingly.
6. Some classes have very large max errors, implying the inconsistent performance of the PCA on different classes of cars.

- **Conclusion**

This week, we implemented PCA as an approach for estimating vehicle orientation using the distribution of edge pixels. The method was first tested on a single vehicle crop and then applied to 240 crops from five aerial images. The results showed that PCA can estimate the main axis of vehicles, although its accuracy varies across vehicle classes. The analysis also showed that the distribution of edges within a crop can influence the estimated orientation. PCA provides another approach to vehicle orientation estimation, which can now be compared with the Radon method.

1. Work done this week

- Studied the working principle of Principal Component Analysis (PCA).
- Implemented PCA to estimate the principal axis of a vehicle from its Canny edge pixels.
- Tested the method on a single vehicle crop and visualised the estimated axis.
- Applied the method to 240 vehicle crops from five aerial images.
- Calculated the orientation estimation errors and analysed their overall distribution.
- Compared mean errors across different vehicle classes.
- Saved the individual results and per-class error statistics in CSV files.

2. Work to be done next week

- Compare the PCA and Radon methods using the same 240 vehicle crops.
- Calculate and compare the overall errors of both methods.
- Compare their performance across different vehicle classes.
- Inspect selected crops with large errors to understand the reasons behind inaccurate estimates.

3. Challenges faced

- We initially found it difficult to understand how eigenvalues and eigenvectors are used in PCA to find the vehicle's main direction and convert the resulting angle into the required 0°–90° range.
- We faced some issues while matching the aerial images with their annotation files and extracting all 240 vehicle crops correctly.
- We also found it challenging to understand why PCA gave different error values for different vehicle classes, especially why some classes had much higher mean errors than others.